

Math 252 - Calculus II - Final Exam

CRN: 23526

March 18, 2025

Last Name: _____, First Name: _____

1. There are 17 problems. Some problems have multiple parts. The total credit is 100.
2. No collaboration is allowed for this exam. The work on this exam is to be yours and yours alone. Failure to adhere to this policy will result in a zero on this exam. Any instances of suspected cheating will also be reported to the Dean of Students' Office.
3. No calculators, books or other material are permitted.
4. In order to receive full credit you need to show your work, and present your answers as clearly as possible. Final answers without work may not receive any credit, unless a problem is a fill-in-blank or multiple choice problem.
5. Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless the problem specifically indicates otherwise.
6. Simplify where appropriate. Quantities like $\sqrt{9}$, $\ln e$, inverse trigonometric functions at special values, or trigonometric functions at special angles should be calculated. Reduce all fractions. Answers that are not simplified may result in loss of points.
7. If an answer line is provided for the answer, place your final answer in it. Otherwise, circle, box, or underline each final answer as appropriate for clarity.
8. Failure to follow directions specific to a problem will result in loss of points.
9. If you work a problem in two different ways, please clearly indicate which one you want us to grade. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
10. Put away and **silence** your cell phones, tablets, laptops, etc.
11. Once this exam begins, you will have 120 minutes to complete your solutions.

1. [2 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^3 - x^2 + 2x + 1}{x(x+2)^3(x^2+4)^2} = \frac{\frac{A}{x} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3} + \frac{Ex+F}{x^2+4} + \frac{Gx+H}{(x^2+4)^2}}$$

2. [2 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

$$\frac{x^3}{\sqrt{x^7-1}} > \frac{x^3}{x^{7/2}} = \frac{1}{x^{1/2}}$$

$p = \frac{1}{2}$

The improper integral $\int_2^{\infty} \frac{x^3}{\sqrt{x^7-1}} dx$ is Divergent (choose "convergent" or "divergent" to fill in the blank) because $\frac{x^3}{\sqrt{x^7-1}} \underline{A} \underline{E}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{7/2}}$, D). $\frac{1}{x^4}$, E). $\frac{1}{x^{1/2}}$, F). $\frac{1}{x^{3/7}}$

3. [5 pts] Evaluate $\int \frac{x^6 - x^3 + 3}{x^4} dx$.

$$\begin{aligned} &= \int (x^2 - \frac{1}{x} + \frac{3}{x^4}) dx \\ &= \frac{1}{3}x^3 - \ln|x| + 3 \cdot \frac{1}{-3} \cdot x^{-3} + C \\ &= \frac{1}{3}x^3 - \ln|x| - \frac{1}{x^3} + C \end{aligned}$$

4. [5 pts] Evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$.

$$= \int_1^2 2e^u du$$

$$= 2e^u \Big|_1^2$$

$$= 2e^2 - 2e$$

$$u = \sqrt{x} \quad du = \frac{1}{2\sqrt{x}} dx$$

$$2du = \frac{1}{\sqrt{x}} dx$$

$$\text{when } x=1, u=\sqrt{1}=1$$

$$\text{when } x=4: u=\sqrt{4}=2$$

5. [6 pts] Evaluate $\int x \sin(5x) dx$.

$$u = x$$

$$dv = \sin(5x) dx$$

$$du = dx$$

$$v = -\frac{1}{5} \cos(5x)$$

$$= -\frac{1}{5} x \cos(5x) + \int \frac{1}{5} \cos(5x) dx$$

$$= -\frac{1}{5} x \cos(5x) + \frac{1}{25} \sin(5x) + C$$

6. [6 pts] Evaluate $\int e^x \sin x \, dx$.

$$u = \sin x$$

$$dv = e^x \, dx$$

Hint: Use integration by parts twice.

$$du = \cos x \, dx$$

$$v = e^x$$

$$= e^x \sin x - \int e^x \cos x \, dx$$

see the work

$$= e^x \sin x - (e^x \cos x + \int e^x \sin x \, dx)$$

$$\Rightarrow \int e^x \sin x \, dx = e^x \sin x - e^x \cos x - \int e^x \sin x \, dx$$

$$2 \int e^x \sin x \, dx = e^x \sin x - e^x \cos x$$

$$\Rightarrow \int e^x \sin x \, dx = \frac{1}{2} e^x \sin x - \frac{1}{2} e^x \cos x + C.$$

$$\int e^x \cos x \, dx:$$

$$u = \cos x$$

$$dv = e^x \, dx$$

$$du = -\sin x \, dx$$

$$v = e^x$$

$$= e^x \cos x + \int e^x \sin x \, dx$$

7. [8 pts] Use the method of partial fractions to evaluate $\int \frac{x+1}{x(x^2+1)} dx$.

Note: No credit will be given if you do not use partial fraction method.

$$\frac{x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$A+B=0$$

$$A=1$$

$$C=1 \Rightarrow B=-1$$

$$A=1 \quad C=1$$

$$x+1 = A(x^2+1) + (Bx+C)x$$

$$x+1 = Ax^2 + A + Bx^2 + Cx$$

$$x+1 = (A+B)x^2 + Cx + A$$

$$\int \frac{x+1}{x(x^2+1)} dx = \int \left(\frac{1}{x} + \frac{-x+1}{x^2+1} \right) dx$$

$$= \int \frac{1}{x} dx - \int \frac{x}{x^2+1} dx + \int \frac{1}{x^2+1} dx$$

↓ work shown below:

$$= \ln|x| - \frac{1}{2} \ln|x^2+1| + \arctan x + C$$

$$u=x^2+1 \quad du=2x dx$$

$$\int \frac{x}{x^2+1} dx = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|x^2+1| + C$$

8. [8 pts] Use trigonometric substitution to evaluate $\int \frac{1}{x^2 \sqrt{x^2 - 4}} dx$.

Note: Simplify your answer so that it does not contain any trigonometric composed with inverse trigonometric terms.

$$x = 2 \sec \theta$$

$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\begin{aligned} \sqrt{x^2 - 4} &= \sqrt{4 \sec^2 \theta - 4} \\ &= \sqrt{4(\sec^2 \theta - 1)} \\ &= \sqrt{4 \tan^2 \theta} \\ &= 2 \tan \theta \end{aligned}$$

$$\int \frac{1}{x^2 \sqrt{x^2 - 4}} dx$$

$$= \int \frac{1}{4 \sec^2 \theta \cdot 2 \tan \theta} 2 \sec \theta \tan \theta d\theta$$

$$= \int \frac{1}{4 \sec \theta} d\theta$$

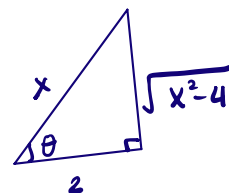
$$= \frac{1}{4} \int \cos \theta d\theta$$

$$= \frac{1}{4} \sin \theta + C$$

$$= \frac{1}{4} \cdot \frac{\sqrt{x^2 - 4}}{x} + C$$

$$x = 2 \sec \theta$$

$$\Rightarrow \sec \theta = \frac{x}{2} = \frac{H}{A}$$



$$\Rightarrow \sin \theta = \frac{\sqrt{x^2 - 4}}{x}$$

9. [8 pts] Find the average value of $f(x) = \sin^2 x \cos^3 x$ on the interval $[0, \frac{\pi}{6}]$.

Note: Simplify your trigonometric values, but you do not need to simplify fractions with distinct denominators if there is any.

$$\begin{aligned}
 f_{\text{Ave}} &= \frac{1}{\frac{\pi}{6}} \int_0^{\frac{\pi}{6}} \sin^2 x \cos^3 x \, dx = \frac{6}{\pi} \int_0^{\frac{\pi}{6}} \sin^2 x \cos^2 x \cos x \, dx \\
 &= \frac{6}{\pi} \int_0^{\frac{\pi}{6}} \sin^2 x (1 - \sin^2 x) \cos x \, dx & u = \sin x, \, du = \cos x \, dx \\
 & & \text{When } x=0, \, u = \sin 0 = 0 \\
 & & x = \frac{\pi}{6}, \, u = \sin \frac{\pi}{6} = \frac{1}{2} \\
 &= \frac{6}{\pi} \int_0^{\frac{1}{2}} u^2 (1 - u^2) \, du \\
 &= \frac{6}{\pi} \int_0^{\frac{1}{2}} (u^2 - u^4) \, du \\
 &= \frac{6}{\pi} \cdot \left(\frac{1}{3} u^3 - \frac{1}{5} u^5 \right) \Big|_0^{\frac{1}{2}} \\
 &= \frac{6}{\pi} \left(\frac{1}{3} \cdot \left(\frac{1}{2}\right)^3 - \frac{1}{5} \left(\frac{1}{2}\right)^5 \right) \checkmark \text{ acceptable answers.} \\
 &= \frac{6}{\pi} \left(\frac{1}{24} - \frac{1}{160} \right) \\
 &= \frac{3}{\pi} \left(\frac{1}{12} - \frac{1}{80} \right) \\
 &= \frac{3}{\pi} \left(\frac{20-3}{240} \right) = \frac{51}{240\pi}
 \end{aligned}$$

10. [8 pts] Evaluate the improper integral $\int_2^{\infty} \frac{1}{x(\ln x)^2} dx$ or show that it is divergent.

Note: Use the limit definition of improper integrals to compute the above integral. Do not use Comparison Test.

$$u = \ln x, \quad du = \frac{1}{x} dx$$
$$\int \frac{1}{x(\ln x)^2} dx = \int \frac{1}{u^2} du = -\frac{1}{u} + C = -\frac{1}{\ln x} + C$$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^2} dx = \lim_{t \rightarrow \infty} \left(-\frac{1}{\ln x} \Big|_2^t \right)$$

$$= \lim_{t \rightarrow \infty} \left(-\frac{1}{\ln t} - \left(-\frac{1}{\ln 2} \right) \right) = 0 + \frac{1}{\ln 2} = \frac{1}{\ln 2}, \quad \text{Convergent.}$$

11. [8 pts] Find the solution to the initial value problem

$$\frac{dy}{dx} = \frac{4x^3}{y^4}, \quad y(0) = 2.$$

Express your answer in the explicit function form of $y = f(x)$.

$$y^4 dy = 4x^3 dx$$

$$\int y^4 dy = \int 4x^3 dx$$

$$\frac{1}{5} y^5 = x^4 + C$$

$$\text{When } x=0, y=2 \Rightarrow \frac{1}{5} \cdot 32 = 0 + C \Rightarrow C = \frac{32}{5}$$

$$\Rightarrow \frac{1}{5} y^5 = x^4 + \frac{32}{5}$$

$$\Rightarrow y^5 = 5x^4 + 32$$

$$y = \sqrt[5]{5x^4 + 32}$$

12. [7 pts] A thin metal bar with a density function of $\rho(x) = 6 + \frac{8}{x^3}$ (in kg/m) lies on the x -axis between $x = 1$ and $x = 2$ (in meters).

1). Find the mass of the bar.

2). Find the center of mass of the bar.

Note: Simplify your answers whenever possible.

$$\begin{aligned} 1). \quad m(x) &= \int_1^2 \rho(x) dx = \int_1^2 \left(6 + \frac{8}{x^3} \right) dx = \left(6x + 8 \cdot \frac{1}{-2} x^{-2} \right) \Big|_1^2 \\ &= \left[12 - 4 \cdot \left(\frac{1}{2} \right)^2 \right] - [6 - 4] \\ &= [12 - 1] - 2 = 9 \end{aligned}$$

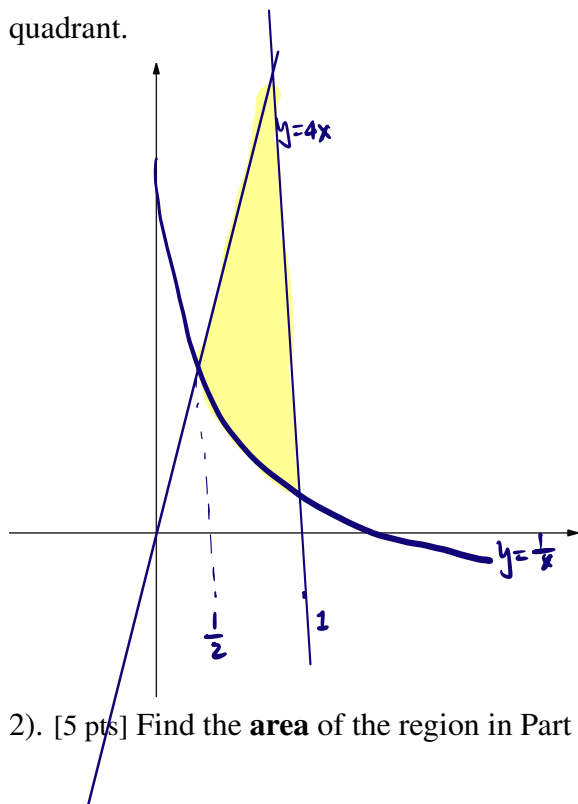
$$\begin{aligned} 2). \quad \int_1^2 x \rho(x) dx &= \int_1^2 x \left(6 + \frac{8}{x^3} \right) dx = \int_1^2 \left(6x + \frac{8}{x^2} \right) dx \\ &= \left(3x^2 - \frac{8}{x} \right) \Big|_1^2 = \left(12 - \frac{8}{2} \right) - \left(3 - 8 \right) = (12 - 4) + 5 = 13 \end{aligned}$$

$$\Rightarrow \bar{x} = \frac{13}{9}$$

Answer: 1). The mass of the bar $m = \underline{9}$ kg.

2). The center of mass of the bar $\bar{x} = \underline{\frac{13}{9}}$ m.

13. [2 pts] 1). Sketch the graph of the region bounded by $y = \frac{1}{x}$, $y = 4x$ and $x = 1$ in the first quadrant. Then find the coordinates of the intersection point of $y = \frac{1}{x}$ and $y = 4x$ in the first quadrant.

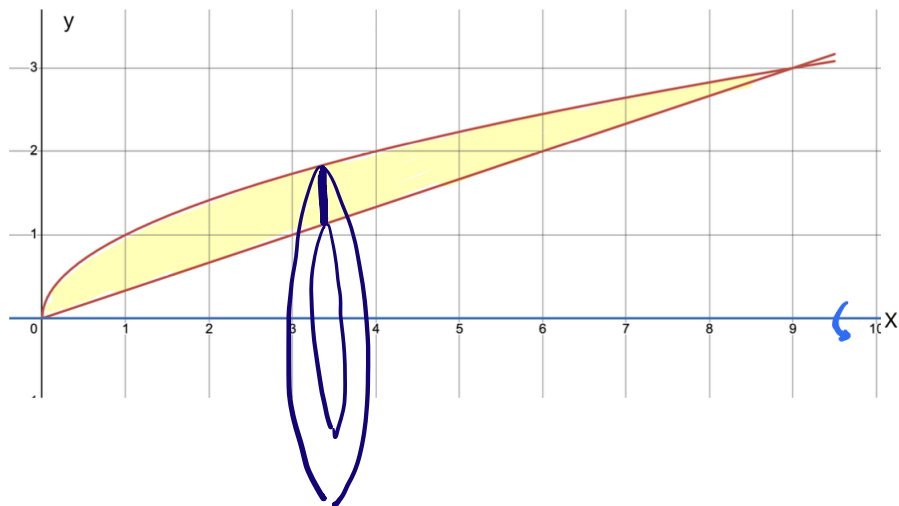


$$\begin{cases} y = 4x \\ y = \frac{1}{x} \end{cases} \Rightarrow \begin{aligned} 4x &= \frac{1}{x} \\ 4x^2 &= 1 \\ x^2 &= \frac{1}{4} \end{aligned} \Rightarrow x = \frac{1}{2}$$

- 2). [5 pts] Find the **area** of the region in Part 1).

$$\begin{aligned} \text{Area} &= \int_{\frac{1}{2}}^1 \left(4x - \frac{1}{x} \right) dx \\ &= \left(2x^2 - \ln|x| \right) \Big|_{\frac{1}{2}}^1 \\ &= (2 - \ln 1) - \left(2 \cdot \left(\frac{1}{2} \right)^2 - \ln\left(\frac{1}{2} \right) \right) \\ &= 2 - \frac{1}{2} + \ln\left(\frac{1}{2} \right) \\ &= \frac{3}{2} + \ln\left(\frac{1}{2} \right) \end{aligned}$$

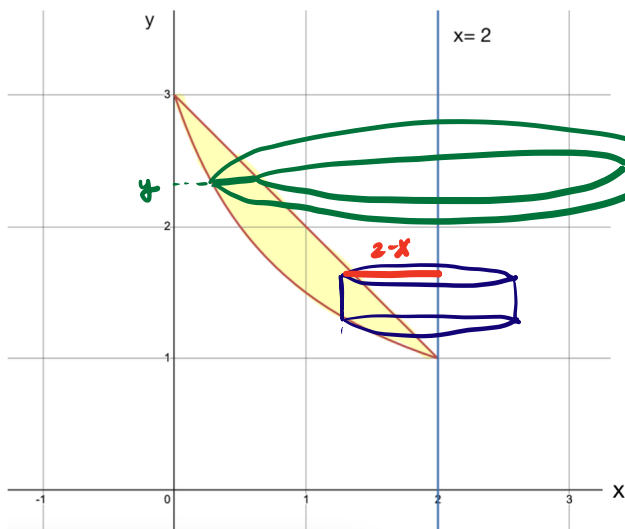
14. [6 pts] The following shaded region is bounded by $y = \frac{x}{3}$ and $y = \sqrt{x}$. Find the **volume** of the solid of revolution obtained by rotating this region about x -axis.



Washer Method:

$$\begin{aligned}
 V &= \pi \int_0^9 \left[(\sqrt{x})^2 - \left(\frac{x}{3}\right)^2 \right] dx \\
 &= \pi \int_0^9 \left(x - \frac{x^2}{9} \right) dx \\
 &= \pi \cdot \left(\frac{1}{2} x^2 - \frac{x^3}{27} \right) \Big|_0^9 \\
 &= \pi \left(\frac{81}{2} - \frac{9 \cdot 9 \cdot 9}{27} \right) \\
 &= \pi \left(\frac{81}{2} - 27 \right) \\
 &= \pi \cdot \frac{27}{2}
 \end{aligned}$$

15. [6 pts] The following shaded region is bounded by $y = 3 - x$ and $y = \frac{3}{x+1}$.



Set up, **but do not evaluate**, definite integrals to compute of the volume of the solid obtained by rotating the above shaded region about the vertical line $x = 2$ using both disk/washer and cylindrical shell methods.

Fill in the blanks in the following answer line. You do not need to simplify the integrand functions or evaluate the above integrals.

Hint: The curve $y = \frac{3}{x+1}$ is equivalent to $x = \frac{3}{y} - 1$.

i). Disk/Washer Method:

$$\text{Volume of the Solid} = \int_{\underline{1}}^{\underline{3}} \frac{\pi [2 - (\frac{3}{y} - 1)]^2 - \pi [2 - (3 - y)]^2}{\underline{\hspace{1cm}}} d\underline{y}$$

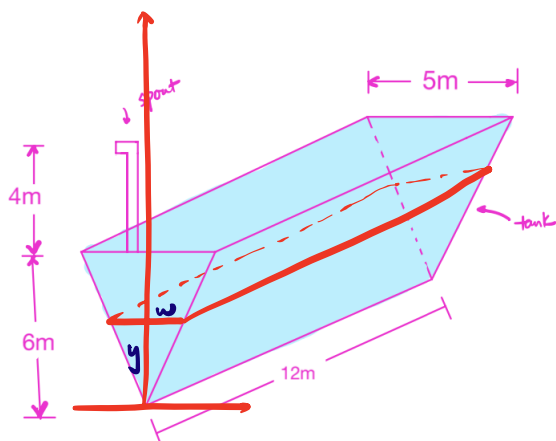
ii). Cylindrical Shell Method:

$$\text{Volume of the Solid} = \int_{\underline{0}}^{\underline{2}} \frac{2\pi (2 - x) (3 - x - \frac{3}{x+1})}{\underline{\hspace{1cm}}} d\underline{x}$$

16. [4 pts] A tank in the following inverted triangular prism shape is *full of water*. Set up a definite integral to compute the work required to pump all the water out of the spout, which is 4m above the top of the tank as shown in the picture.

The density of the water is 1000 kg/m^3 , and the gravitational acceleration is 9.8 m/s^2 .

Fill in the blanks in the following answer line. **Do not evaluate the definite integral.**



$$\frac{w}{y} = \frac{5}{6}$$

$$w = \frac{5}{6}y$$

$$A(y) = \frac{5}{6}y \cdot 12 = 10y$$

Answer: Work = $\int_0^6 1000 \cdot 9.8 (10y)(10-y) dy$ J.

17. [4 pts] A population of squirrels (in hundreds) that lives in a forest is modeled by the following logistic equation with the initial population,

$$\frac{dP}{dt} = \frac{1}{150}P(30 - P), \quad P(0) = 6.$$

- For what values of P is the population increasing? Answer: $P < 30$
- For what values of P is the population decreasing? Answer: $P > 30$
- Find the population function $P(t)$, that is, the solution to the logistic equation.

Answer: $P(t) = \frac{30}{1 + 4e^{-0.2t}}$

- Find $\lim_{t \rightarrow \infty} P(t)$. Answer: 30

$$\frac{dP}{dt} = \frac{30}{150} P \left(1 - \frac{P}{30} \right) = 0.2 P \left(1 - \frac{P}{30} \right)$$

$$K=30, \quad r=0.2 \quad P_0=6$$

$$\Rightarrow A = \frac{K-P_0}{P_0} = \frac{30-6}{6} = 4$$

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\int \cot x \csc x dx = -\csc x + C, \quad \int \csc^2 x dx = -\cot x + C$$

$$\int \tan x dx = \ln |\sec x| + C, \quad \int \cot x dx = \ln |\sin x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C, \quad \int \csc x dx = \ln |\csc x - \cot x| + C$$

$$\int \sec^3 x dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases} \quad \int_0^1 \frac{1}{x^p} dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

$$m = \int_a^b \rho(x) dx, \quad \bar{x} = \frac{\int_a^b x\rho(x) dx}{\int_a^b \rho(x) dx}, \quad W = \int_a^b F(x) dx, \quad W = \int_0^b \rho g A(y)(h-y) dy$$

$$\text{Solutions to } \frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right), P(0) = P_0 \text{ is } P(t) = \frac{K}{1 + Ae^{-rt}}, \text{ where } A = \frac{K - P_0}{P_0}.$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined